

aygun.varol@tuni.fi | Web: aygunvarol.github.io | GitHub: [aygunvarol](https://github.com/aygunvarol)
LinkedIn: [aygunvarol](#) | ORCID: [0000-0002-4029-7676](https://orcid.org/0000-0002-4029-7676) | Google Scholar

AI/IoT Research Engineer with 5+ years of experience in developing smart environments through integration of Internet of Things (IoT) and Artificial Intelligence (AI). Expertise in Large Language Model (LLM) deployment on edge devices, AI security research, and end-to-end IoT system design.

DEGREES

D.Sc., Computing and Electrical Engineering, GPA: 4.71/5.0
[Tampere University, Tampere, Finland, 2023 — Present](#)

M.Sc., Electrical and Electronics Engineering, GPA:3.8/4.0
[Isparta University of Applied Sciences, Isparta, Türkiye, 2020 — 2022](#)

B.Sc., Electrical and Electronics Engineering, GPA:3.35/4.0
[Manisa Celal Bayar University, Manisa, Türkiye, 2013 — 2017](#)

OTHER EDUCATION AND EXPERTISE

The European Summer School on Artificial Intelligence (ESSAI & ACAI)
[National and Kapodistrian University of Athens, Greece, 2024](#)

LANGUAGE SKILLS

Turkish, [Native Language](#)

English, [Proficient \(C1\), IELTS: 7](#)

Finnish, [Elementary](#)

German, [Elementary](#)

CURRENT EMPLOYMENT

Doctoral Researcher

[Faculty of Information Technology and Communication Sciences, Tampere University](#)
[Tampere, Finland, August 2023 — Present](#)

My doctoral research focuses on AI-driven smart indoor environments, IoT, and the risks of deploying autonomous AI agents in everyday settings. I work as a doctoral researcher in the EVIL-AI project (*Identification and Mitigation of the Negative Effects of Artificial Intelligence Agents*), a multidisciplinary research project funded by the Jane and Aatos Erkko Foundation.

PREVIOUS WORK EXPERIENCE

Research and Teaching Assistant

[Faculty of Technology, Isparta University of Applied Sciences](#)
[Isparta, Türkiye, January 2020 — December 2023](#)

Electrical Maintenance Engineer Intern

[Habaş Industrial and Medical Gases Production Industries Inc.](#)
[İzmir, Türkiye, July 2016](#)

RESEARCH FUNDING AND GRANTS

Study Abroad Programme Scholarship

Funding Body: Republic of Türkiye Ministry of National Education

Duration: 2023 – 2027 (4 years)

Budget: 70,000€

Role: Doctoral Scholarship Recipient

Activity: Competitive programme sponsoring Turkish students for postgraduate studies abroad.

RESEARCH OUTPUT

Publications

13 Publications	69 Citations	4 h-index	2 i10-index
--------------------	-----------------	--------------	----------------

List of Publications

1. **Varol, A.**, Shaikh, A., Motlagh, N. H., Leino, M., & Virkki, J. (2026, August). Plot-and-Ask: Multimodal Local LLMs in Smart Environments for Visual IoT Analytics. In *Proceedings of the 21st Conference on Computer Science and Intelligence Systems (FedCSIS)* (ACSIS, Vol. 47, pp. 609–614). PTI.
Conference Proceedings
2. **Varol, A.**, Shaikh, A., Motlagh, N. H., Leino, M., & Virkki, J. (2026, June). Multi-Agent Large Language Models for Distributed Internet of Things Analytics in Smart Environments. In *2026 11th International Conference on Smart and Sustainable Technologies (SpliTech)*. IEEE.
Conference Proceedings
3. Savolainen, I., Vestelin, J., Ihalainen, T., **Varol, A.**, & Virkki, J. (2026, June). A Structured Prompt-Based Approach to Evaluating Exploitative Misuse in Generative AI. In *2026 11th International Conference on Smart and Sustainable Technologies (SpliTech)*. IEEE.
Conference Proceedings
4. Vuohijoki, T., Huhtasalo, J., Raukola-Lindblom, M., Deloso, R., Sakif, M. M., Kivin, O., Rautio, T., Tavakoli, A., Gonzalez Visiedo, C., Mattila, H., Rahman, S. M., Sipola, P., Ihalainen, T., **Varol, A.**, Virkki, J., & Merilampi, S. (2026, June). User-Centred Concept Creation of Cloth-Based Activators to Support Social Inclusion. In *2026 11th International Conference on Smart and Sustainable Technologies (SpliTech)*. IEEE.
Conference Proceedings
5. **Varol, A.**, Shaikh, A., Jalo, H., Muchenje, G., de Cerqueira, J. S., Kemell, K.-K., Motlagh, N. H., Leino, M., Pirkkalainen, H., Abrahamsson, P., & Virkki, J. (2026). Evil-AI Benchmark: An Evaluation Framework for Adversarial Security Assessment of LLM Agents in Smart Environments. *IEEE Access*, 14, 133098–133116.
Journal
6. Motlagh, N. H., **Varol, A.**, Zaidan, M. A., Fung, P. L., Varjonen, S., Nurmi, P., & Tarkoma, S. (2026). Measuring the Atmosphere: Environmental Sensing for Occupant Flows in Commercial Spaces. *IEEE Internet of Things Journal*.

Journal

7. **Varol, A.**, Motlagh, N. H., Leino, M., Tarkoma, S., & Virkki, J. (2026). Creation of AI-driven smart spaces for enhanced indoor environments – a survey. *Internet of Things*, 36, 101876.

Journal

8. **Varol, A.**, Motlagh, N. H., Leino, M., & Virkki, J. (2025, June). Performance of Large Language Models Across Edge and Cloud Platforms in Smart Spaces. In *2025 10th International Conference on Smart and Sustainable Technologies (SpliTech)* (pp. 1–6). IEEE.

Conference Proceedings

9. Shaikh, A., **Varol, A.**, & Virkki, J. (2025). From Prompts to Motors: Man-in-the-Middle Attacks on LLM-Enabled Vacuum Robots. *IEEE Access*, 13, 137505–137513.

Journal

10. **Varol, A.**, Yucel, F., Yuce, E., & Cakir, A. (2024). A single-IC realizable, electronically tunable, OTA-based full-wave rectifier with simultaneous positive and negative outputs. *AEU – International Journal of Electronics and Communications*, 183, 155374.

Journal

11. Kayaalp, K., & **Varol, A.** (2024). LeNet ve ResNet Derin Öğrenme Modelleri ile Asma Yapraklarının Sınıflandırılması. *Veri Bilimi*, 7(1), 16–25.

Journal

12. **Varol, A.**, Yucel, F., & Cakir, A. (2023). A new electronically tunable transimpedance-mode OTA-based first-order universal filter and its quadrature oscillator application. *Journal of Circuits, Systems and Computers*, 32(11), 2350184.

Journal

13. **Varol, A.**, & Çetin, B. K. (2020). A New Mobile Application For Physical Measurement in A Cellular Network. *Journal of Scientific Reports-A*, (045), 178–200.

Journal

Preprints

1. **Varol, A.**, Kołodziej, K., Sobczak, Ł., Romaszewski, M., Głomb, P., Motlagh, N. H., Leino, M., & Virkki, J. (2026). Enabling Cloud-Level Accuracy in Edge AI through IoT Data Preprocessing. *arXiv preprint arXiv:2606.22496*.

Preprint

RESEARCH SUPERVISION AND LEADERSHIP

Graduate Supervision

Amber Ikram	Master's Student	Tampere University	Co-Supervisor 2025 — Present
-------------	------------------	--------------------	------------------------------

Undergraduate Supervision

Level	Role	Institution	Period
Blaise Mwubahamana (Bachelor's)	Supervisor	Tampere University	2026–Present
Bachelor's (5 projects)	Supervisor	Isparta University of Applied Sciences	2020–2023

Isparta undergraduate projects supervised under the TÜBİTAK 2209-A Research Grant Program for University Students.

TEACHING MERITS

Teaching Assistant

DPHAT 100 | [Gameful Realities Doctoral Seminar](#)
Tampere University, 2025 — 2026

EEM-203 | [Electronics I](#)
Isparta University of Applied Sciences, 2020 — 2024

EEM-209 | [Digital Systems](#)
Isparta University of Applied Sciences, 2020 — 2024

EEM-204 | [Electronics II](#)
Isparta University of Applied Sciences, 2019 — 2023

EEM-252 | [Digital System Design](#)
Isparta University of Applied Sciences, 2019 — 2023

EEM-301 | [Microprocessor System Design](#)
Isparta University of Applied Sciences, 2021 — 2023

EEM-114 | [Fundamentals of Electrical and Electronics](#)
Isparta University of Applied Sciences, 2019 — 2022

EEM-202 | [Circuit Analysis II](#)
Isparta University of Applied Sciences, 2020 — 2021

Guest Lecturer

HTI.470 | [Human-Centred and Societal Perspectives to Artificial Intelligence](#)
Tampere University, Tampere (Remote), 2 February 2026

OTHER ACADEMIC MERITS

Reviewing

[Journal, IEEE Access](#)
External Reviewer | June 2026

[Conference, Nordic Conference on Human-Computer Interaction \(NordiCHI 2026\)](#)
External Reviewer | June 2026

[Journal, Internet of Things, ISSN: 2542-6605](#)
External Reviewer | May 2026

[Conference, The 35th IEEE International Conference on Robot and Human Interactive Communication \(RO-MAN 2026\)](#)
External Reviewer | April 2026

Chairing

[SpliTech 2026 – 11th International Conference On Smart And Sustainable Technologies](#)
Academic Chair, Split, Croatia | June 2026

[EnvSys 2026 – 14th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems](#)
Academic Chair, Saint Malo, France | May 2026

[EnvSys 2025 – 13th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems](#)
Academic Chair, Irvine, USA | May 2025

Invited Talks

AI Agents in Smart Spaces

The 1st Winter School on 3D Digitainability, Tampere University, Tampere | December 2024

Presentations

Plot-and-Ask: Multimodal Local LLMs in Smart Environments for Visual IoT Analytics

FedCSIS 2026, Riga, Latvia | August 2026

Generative AI Risks and Where to Find Them

Tampere University, Tampere | September 2025

Performance of Large Language Models Across Edge and Cloud Platforms in Smart Spaces

SpliTech 2025, Split, Croatia | June 2025

Risks of Employing AI Agents in Smart Environments

Tampere University, Tampere | May 2025

OTHER MERITS

Technical Competencies

Internet of Things

End-to-end IoT prototyping (device, data collection, processing, dashboards/UIs)

Artificial Intelligence & Machine Learning & Large Language Models

Machine learning, deep learning training, large language models fine-tuning, quantization, deployment, across edge devices and cloud platforms; both proprietary and open-source models

Local & Embedded Systems

Mac Mini, Raspberry Pi, Arduino, edge devices AI and sensor deployments, time-series data analytics, monitoring and automation

Sensor Networks

Wireless communication technologies and protocols, Wi-Fi, HTTP, REST

Developer Tools

Visual Studio, Android Studio, MATLAB, PSpice, Origin

Programming Languages

Python, C, C++, Java, $\text{\LaTeX}$